SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID – 25039**
- **Problem Statement Title- Integrated Platform for Crowdsourced Ocean Hazard Reporting and Social Media Analytics**
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID- T080**
- **Team Name - Team ResQ**

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Problem Statement

India's vast coastline is exposed to **hazards like tsunamis, storm surges, and abnormal sea behavior**. Though agencies like INCOIS provide satellite and sensor-based warnings, real-time citizen and community field reporting is often delayed. Insights from social media during hazard events remain untapped, despite their potential to **reveal ground realities, public awareness, and information spread.**

Solution Approach

Citizen Reporting

Citizens can instantly report hazards such as high tides, cyclones, or oil spills. Reports include photos, videos, and GPS location to ensure accuracy.

AI Verification

AI-powered models automatically filter out spam reports, analyze media content for authenticity, and assign a confidence score. This ensures only credible reports reach authorities, reducing false alarms.

Samudram

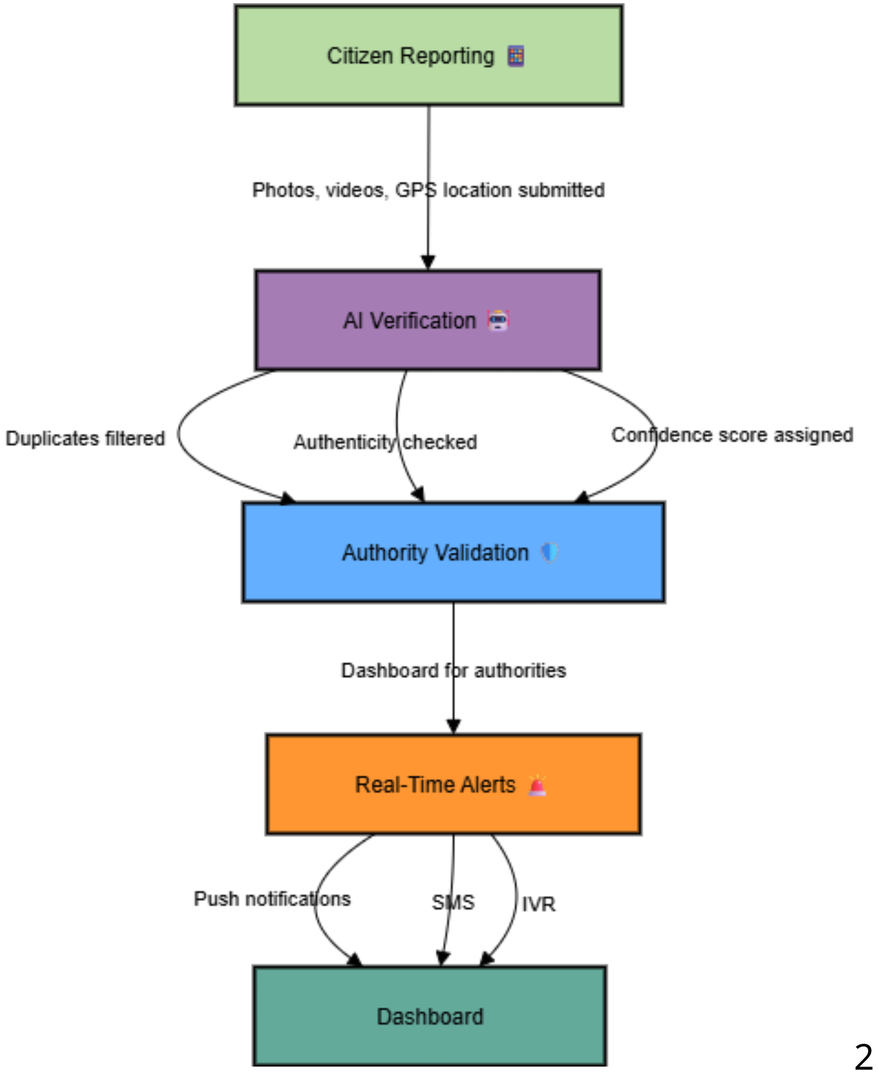
Authority Validation

A dedicated dashboard enables authorities to view, validate, and act on incoming reports. They can mark hazards as Pending, Verified, or Resolved, and escalate critical alerts to higher agencies for rapid action

Real-Time Alerts

Once validated, instant alerts are sent to nearby citizens and coastal communities through push notifications. Alerts are color-coded for quick understanding and immediate response.

Data Flow



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Technical Workflow

User

User is allowed to carry out two processes

Login

Alert

Authority

Citizen

Authority

Citizen

Authority Dashboard

Authority has the power to verify the alerts and also define its severenity

Citizen Dashboard

Citizen can have insights on information about disasters , clusters , and safety guidelines for particular disasters

Alerts provided by Authority will directly get verified and move to disaster dashboard

Alerts provided by citizens will be stored in database for further verification

Alerts will contain images,comments,location

Database

Database contains all signedUp citizens ,authority and all unverified alerts

Alert

Alert button on Authority Dashboard

Alert

Alert button on Citizen Dashboard

Disaster Dashboard

contains verified disasters on coastline

Data will not only taken from crowd it will also be sourced from social media sites using Web Scraping

Web Scrabbed Data

Verified clusters will contain several alerts and these alerts will be again verified by authorities for security

ML Framework

Clustering Model	Clusters alerts from same location into a hotspots
NLP	Understands text and comments from each cluster
Images	Recognize and analyze images from each cluster

Output produced by ML Framework

The ML Framework will give the verified clusters and each cluster contains several alerts which further contains images,text,location

Notification will be sent regarding verified clusters alongwith safety measures

Verified Cluster

For unverified clusters Authorities will be notified to see if there is anything serious

Unverified Cluster

Notification

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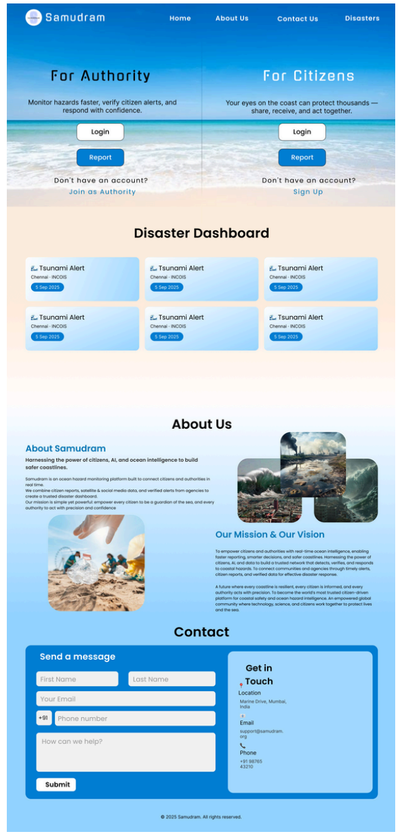
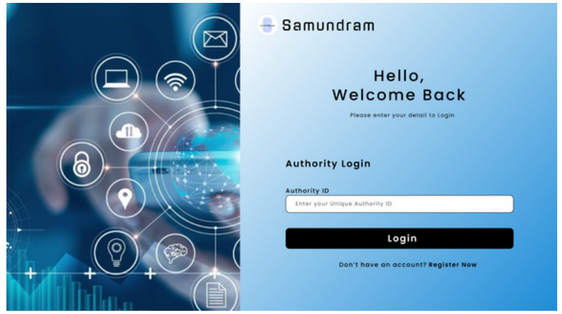
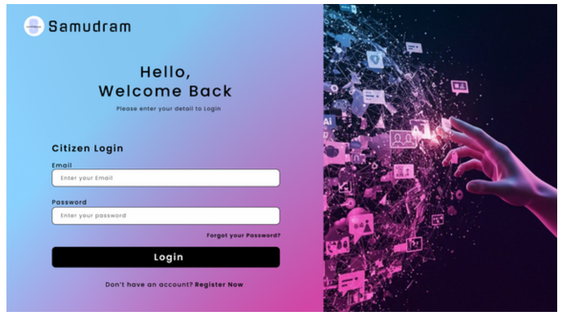
Prototype

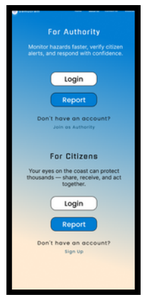


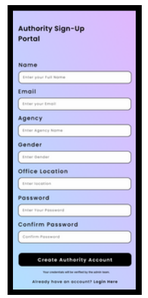
Prototype Link :-

https://sih-frontend-two.vercel.app









User Mobile UI

Feasibility

Economic Feasibility

Low-cost mobile/web app with cloud backend reduces infrastructure needs.

Operational Feasibility

Citizens can easily report hazards via a simple mobile/web interface.

Scalability

Modular design enables integration with future IoT sensors, or satellite data.

Sustainability

Designed for long-term use by integrating with INCOIS systems.

Technological Feasibility

Uses existing, proven technologies (mobile apps, cloud hosting, geospatial databases, NLP models).

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Tech Stacks

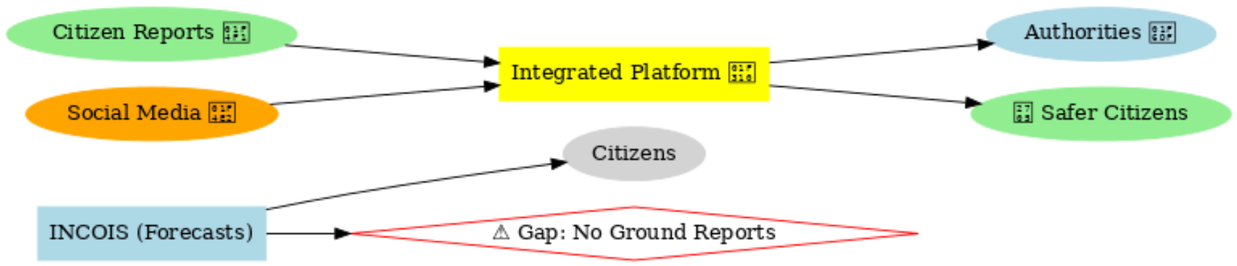
Impact and Benefits

Enhanced Disaster Preparedness – Real-time field data and social media analytics improve accuracy of early warnings.
Faster Emergency Response – Quick validation and situational awareness reduce delays in action.
Community Involvement – Citizens become active contributors in hazard reporting.
Scalable National Solution – Can be extended across coastal regions for long-term resilience.

Future Scope

Citizen reporting with social media analytics.
Integration of IoT sensors & drones for real-time hazard monitoring.
Blockchain-based validation for trusted citizen reports.
. **Global expansion** with cross-border coastal monitoring.
Policy support for data-driven disaster management.

Data flow



RESEARCH AND REFERENCES

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3. IEEE Xplore – Social Media Analytics for Disaster Management

4. Ushahidi Crisis Mapping Platform: https://www.ushahidi.com

5. National Disaster Management Authority (NDMA): https://ndma.gov.in

6. Research Article: Leveraging Social Media for Natural Disaster Management – Springer, 2022

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